

Task 2.2: Regional Analysis Report

Geographic and Geospatial Analysis — National Electricity Grid Network

Prepared by: Member 3 (Visualization Specialist)

1. Overview

This report looks at where Ghana's electricity substations and power lines are located. We look at which regions have more or fewer substations, how Ghana connects to nearby countries through the West African Power Pool (WAPP), and where the grid might be weak or risky. The findings come from three things: counting substations by region, measuring line lengths, and grouping substations that are close together on a map. Charts and the map ([grid_map.html](#), [substation_density_by_region.png](#), [distance_distribution.png](#), [substation_clustering.png](#)) go along with this report.

2. How Many Substations Are in Each Region

Some regions have a lot of substations, and some have very few:

Region	Substations	Total Capacity (MVA)	Avg. Voltage (kV)
Greater Accra	6	520.2	192.3
Ashanti	5	1,099.6	132.4
Volta	4	470.3	90.8
Central	4	316.2	105.3
Eastern	4	828.5	54.5
Western	4	323.0	51.0

Bono	3	252.6	132.0
Northern	3	155.6	124.7
Upper East	2	520.7	161.0
Upper West	1	27.1	11.0

Greater Accra and Ashanti have the most substations. This makes sense — both have big cities (Accra and Kumasi) that need a lot of electricity. Ashanti has slightly fewer substations than Greater Accra, but its substations together can handle more power (1,099.6 MVA total). This means Ashanti has fewer but bigger substations.

The number of substations gets smaller as you go further north. This matches the fact that northern Ghana has fewer people and less industry. One region is a bigger problem than the rest — we look at that in Section 4.

The WAPP connection points are different. These are not regular regions — each one is a single connection point to a neighboring country, so it's normal for each to only have one substation:

Node	Region	Voltage (kV)	Capacity (MVA)
Bolgatanga Interconnection	Burkina Faso border	330	156.1
Elubo Border Station	Cote d'Ivoire border	330	285.1
Aflao Border Station	Togo border	330	423.2
Lome Transmission Hub	Togo	161	120.2
Cotonou Transmission Hub	Benin	161	487.6
Abidjan Transmission Hub	Cote d'Ivoire	161	262.6
Bobo-Dioulasso Hub	Burkina Faso	161	445.9
Conakry Transmission Hub	Guinea	161	251.6

The three stations right on Ghana's border (Bolgatanga, Elubo, Aflao) all run at 330kV, the highest voltage in the dataset. This tells us these three are Ghana's main gateways to the regional power grid. The hubs further inside other countries run at a lower voltage (161kV).

3. Line Length Analysis

We sorted the power lines into three groups by length: short (under 50km), medium (50–150km), and long (over 150km):

Category	Count
Short	15
Medium	28
Long	12

There's a clear pattern: **11 of the 12 "Long" lines run at 330kV** — the highest voltage. This tells us the long-distance lines are the ones doing the heavy, long-haul work of moving power across the country. The lower-voltage lines (11/33/69kV) are mostly short or medium — these are the local lines that deliver power to nearby towns.

We also checked the straight-line distance between each pair of substations and compared it to the listed line length. The listed lengths were about 13–15% longer than the straight-line distance, for every line. This makes sense — real power lines don't travel in a perfectly straight line, they follow roads and land. So this isn't an error in the data, it's just how real transmission lines work. (See [distance_distribution.png](#) for more detail.)

4. Weak Spots in the Grid

The Upper West is the weakest region. It only has one substation, the lowest total power capacity of any region (27.1 MVA), and its one substation runs at the lowest voltage (11kV, local use only — no big transmission line). Upper East and Northern also have fewer substations than most regions, but Upper West stands out as the clearest problem.

There's a second, related issue. In Section 5, we found that **Bolgatanga** (Upper East's main substation) is powerful but sits alone — no other strong substation is nearby. Since Upper East already has very few substations, this is risky: if Bolgatanga stops working, there's no nearby backup to help cover the region.

5. Which Areas Each Utility Covers

The map ([grid_map.html](#)) has a separate layer for each utility, showing only that utility's lines. *(Fill this in after looking at the map: which utility covers which regions, whether a utility's lines are spread out across the country or stay in one area, and whether any utility seems to only serve the WAPP border stations.)*

6. Substations That Cluster Together

We looked at the strongest substations (top 25% by capacity) and checked which ones are close to each other (within 60km):

- **One real cluster was found:** Ejisu, Konongo, and Nkawkaw substations are all close together, near the border of the Ashanti and Eastern regions. This matches what we found earlier — Ashanti has the highest total capacity of any region. This area looks like the main power hub of the network, and having several strong substations close together is good for backup if one fails.
- **8 strong substations are on their own**, with no other strong substation nearby. Most of these (5 of 8) are the WAPP border and hub stations — this is expected, since these points are meant to be spread out to reach different neighboring countries. The other three — **Suhum, Ho, and Bolgatanga** — are regular Ghanaian substations with no backup nearby. Bolgatanga is the most worrying one, since it's also in a region that already has very few substations (see Section 4).

(See [substation_clustering.png](#) for the picture.)

7. Main Findings

1. Regions with bigger cities (Greater Accra, Ashanti) have more substations and more power capacity.
2. Ghana connects to its neighbors mainly through three 330kV border stations: Bolgatanga, Elubo, and Aflao.
3. Longer lines almost always run at the highest voltage (330kV) — these are the backbone of the network.
4. **Upper West** has the fewest substations and the least power capacity of any region — it's the biggest weak spot.
5. **Bolgatanga** is powerful but has no nearby backup, and it's in a region that's already short on substations. This is a risk worth pointing out.

6. **Ejisu, Konongo, and Nkawkaw** (near the Ashanti/Eastern border) form a natural cluster of strong substations, which is good for backup and reliability in that area.